Why do combinatorial optimization problems remain hard even with modern computers?

"For finding the best solution from a finite set of discrete feasible solution, by maximizing or minimizing the objective function."

- Travelling Salesman Problem (finding the optimal delivery routes)
- Max-Cut Problem
- Knapsack Problem (Packing items into containers to maximize the space)

So, what makes this problem difficult ???

Consider the Travelling Salesman Problem (TSP) for 50 cities …

The number of possible routes are

$$N = \frac{(n-1)!}{2}$$

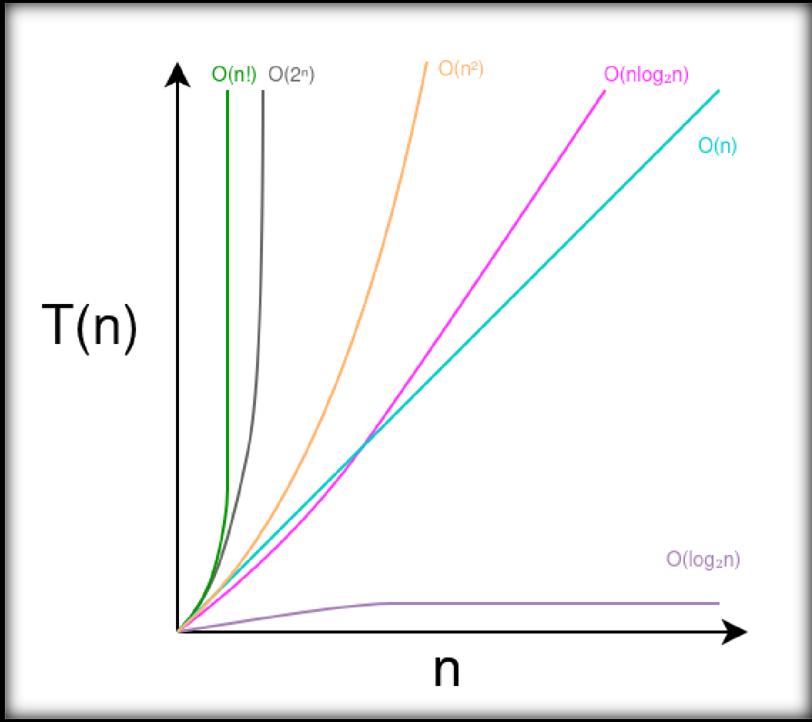
For " n =50 ",

$$\frac{49!}{2} \backsim 3 * 10^{62}$$

"Assume the world's most powerful computer:"

10^{18} *calculations per second*

$$\text{Time} = \frac{3*10^{62}}{10^{18}} \backsim \boxed{3 * 10^{44} sec}$$



Time Complexity Comparison of Prolems

Now here's the interesting part:

" *the age of the universe is* $\boxed{4.35 * 10^{17} sec}$

MIND BOGGLING !!

Is this limitation ***Computational, Mathematical*** *or* ***Physical*** *???*

"The number of possible solution in problems is fixed by **Mathematics.**"

"Faster CPUs do not change exponential growth."

In Classical Computers, physical states are evaluated sequentially - one configuration at a time

*Is this limitation **Computational, Mathematical** or **Physical** ???*

"The number of possible solution in problems is fixed by **Mathematics.**"

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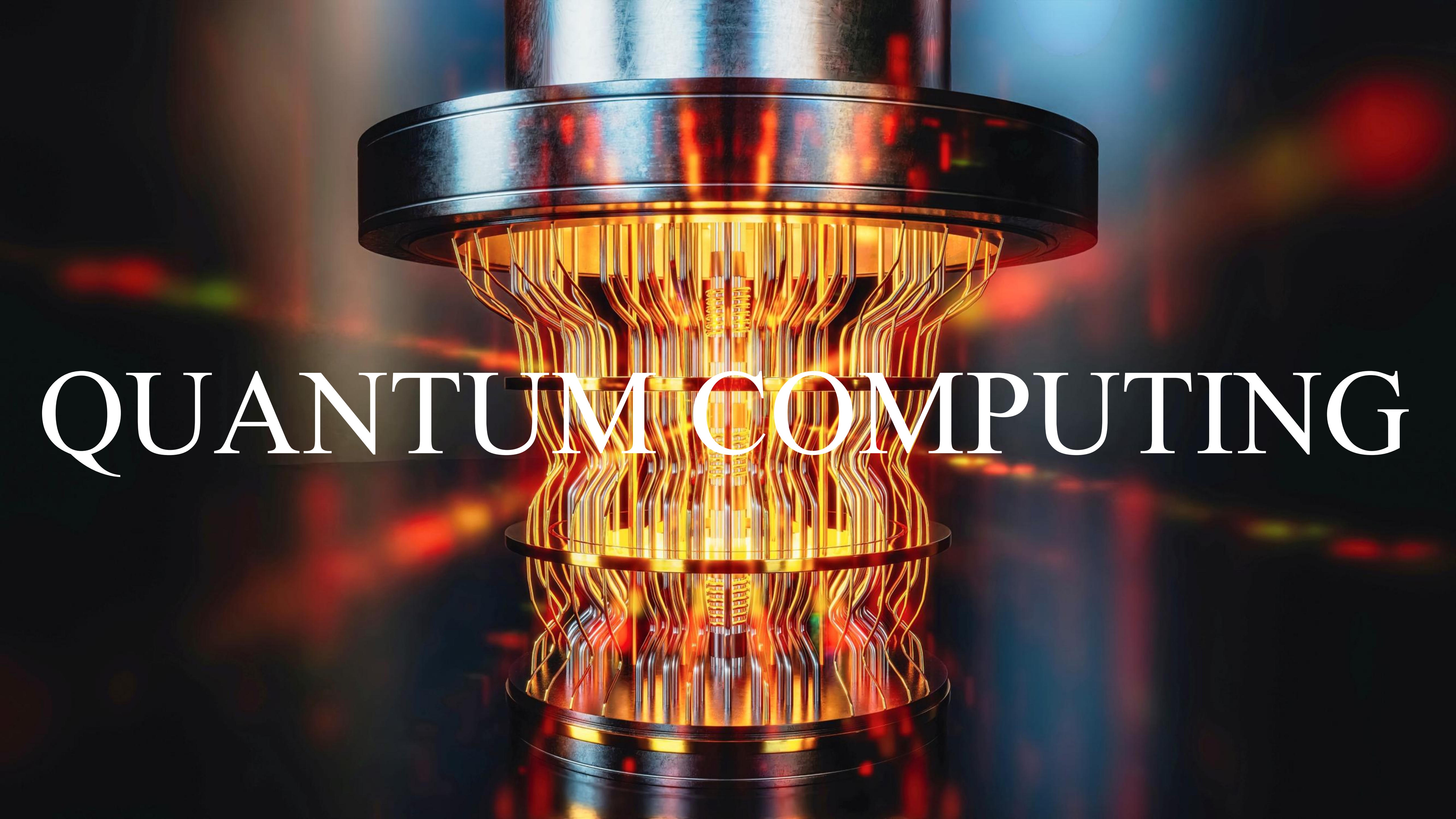
In Classical Computers, physical states are evaluated sequentially – one configuration at a time

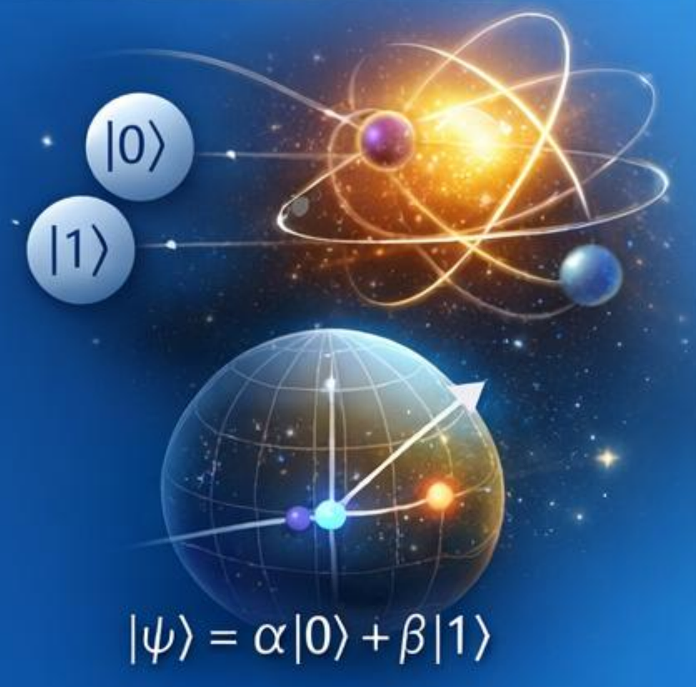
Classical Physics, forces us to explore every possibility one by one. → We need to change physical models of computation.


QUANTUM COMPUTING

Quantum Computing

Quantum States

Uses **quantum states** instead of classical bits to represent information.

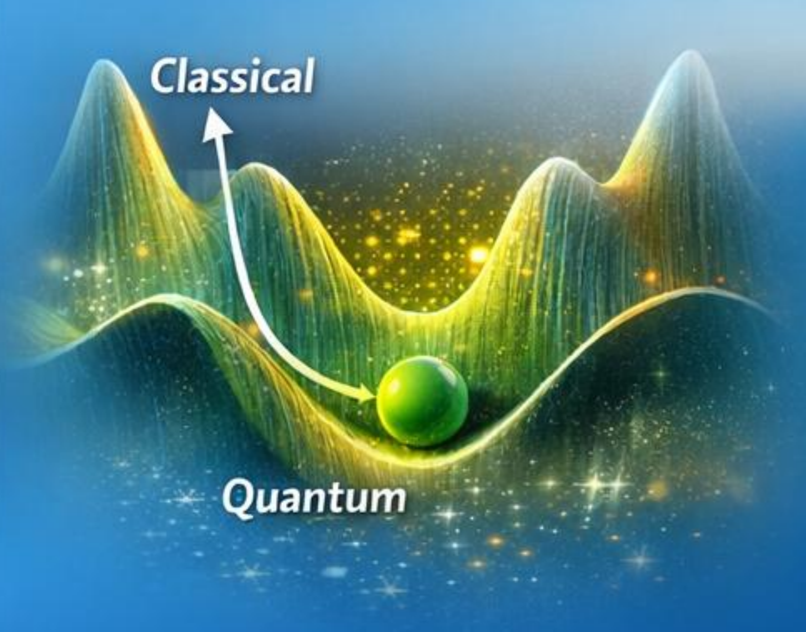


Superposition & Interference

Explore many configurations simultaneously.

Potential Advantage

Can ***outperform*** classical methods on certain problems.



AI-assisted image; original concept by me

Quantum Algorithms for Optimization of Constraint Problems

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Problem Statement

"

This project deals with optimization problems under constraints, which are notoriously hard to solve as system size growth.

"

Goal

"

Explore Quantum Algorithms for tackling such problems.

"

Quantum Algorithms

“ Computation based on Quantum Mechanics Principle to solve problems efficiently and much faster than classical computer . ”

- Shor’s Algorithm (for factoring large numbers)
- Gover’s Algorithm (for searching database)

“Hybrid (Quantum-Classical) Algorithm”

Quantum Approximate Optimization Algorithm (QAOA)

" QAOA is a discretized form of Quantum Adiabatic Computing."

$$U(t) \approx exp\left(-\frac{i}{h}\int_0^t H(\tau)\ d\tau\right)$$

Lie Trotterization,

$$e^{A+B} = \lim_{n\to\infty}\left(e^{A/n}e^{B/n}\right)^n$$

$$\mathrm{U(t)} = \lim_{n\to\infty}\prod_{j=1}^{n}[\ e^{-\frac{i}{h}\{H_j(\mathrm{t})\triangle \mathrm{t}\}}\ ] \qquad , \triangle\, \mathrm{t} = \frac{\mathrm{t}}{\mathrm{n}}$$

$$H(t) = (1\text{-}t)\boldsymbol{H}_{\mathrm{M}} + t\boldsymbol{H}_{\boldsymbol{C}}$$

Easy Hamiltonian($\boldsymbol{H}_{\mathrm{M}}$) - a Hamiltonian whose ground state is easy to prepare
Hard Hamiltonian($\boldsymbol{H}_{\boldsymbol{C}}$) - a Hamiltonian in which we encode the solution of our problem

$$\boldsymbol{U}_{(\boldsymbol{QAOA})}(\boldsymbol{\beta}, \boldsymbol{\gamma}) = \prod_{\boldsymbol{k}=\mathbf{1}}^{\mathrm{p}} \boldsymbol{e}^{(-\boldsymbol{i}\boldsymbol{\beta}_{\boldsymbol{k}}\boldsymbol{H}_{\mathrm{M}})}\, \boldsymbol{e}^{(-\boldsymbol{i}\boldsymbol{\gamma}_{\boldsymbol{k}}\boldsymbol{H}_{\boldsymbol{C}})}$$

$$\beta = \left(1 - \frac{j}{n}\right)\frac{t}{nm}$$

$$\gamma = \left(\frac{j}{n}\right)\frac{t}{nm}$$

How we encode our problem in Hamiltonian?

QUBO Formulation

(Quadratic Unconstrained Binary Optimization)

"QUBO is not a quantum thing, it's just the classical encoding of the optimization problem."

$$\boldsymbol{f(x) = x^T Q\, x = \sum_{i,j}^{n} Q_{ij}\, x_i\, x_j}$$

x = binary variables
Q = real symmetric matrix encoding cost

By mapping binary variables to spin operators

$$x_i = \frac{1 - \widehat{z_i}}{2}$$

$$\boxed{\widehat{H_C} = \mathrm{C} + \sum_i h_i Z_i + \sum_{i<j} J_{ij} Z_i Z_j}$$

$$\left|\psi_{(final)}\right> = U_{(QAOA)}(\beta,\gamma)\left|\psi_{(initial)}\right>$$

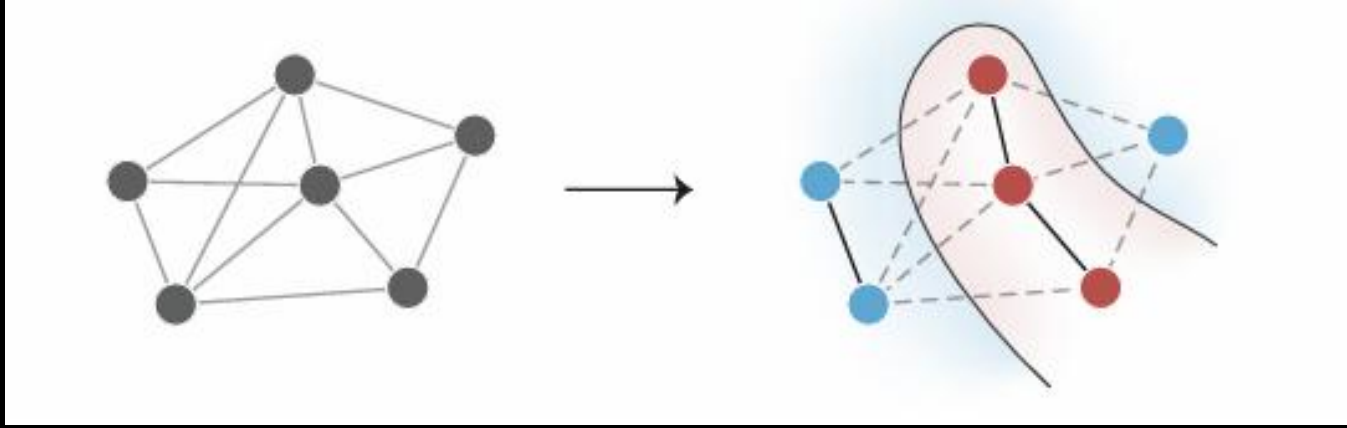
$$|\boldsymbol{\psi}_{(initial)}> = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}} |\boldsymbol{x}>_n$$

$$\boldsymbol{U}_{(QAOA)}(\boldsymbol{\beta},\boldsymbol{\gamma}) = \prod_{k=1}^{n} e^{(-i\beta_k H_{\mathrm{M}})}\, e^{(-i\gamma_k H_C)}$$

$$\boldsymbol{H}_{\mathrm{M}} = \sum_{\mathrm{i}=1}^{\mathrm{n}} \mathrm{X}_{\mathrm{i}}$$

$$\boxed{\left|\boldsymbol{\psi}_{(QAOA)}(\boldsymbol{\beta},\boldsymbol{\gamma})\right> = \prod_{k=1}^{\mathrm{p}} e^{(-i\beta_k H_M)}\, e^{(-i\gamma_k H_C)} \left|\boldsymbol{\psi}_{(initial)}\right>}$$

Max-Cut Problem



The objective function in this case

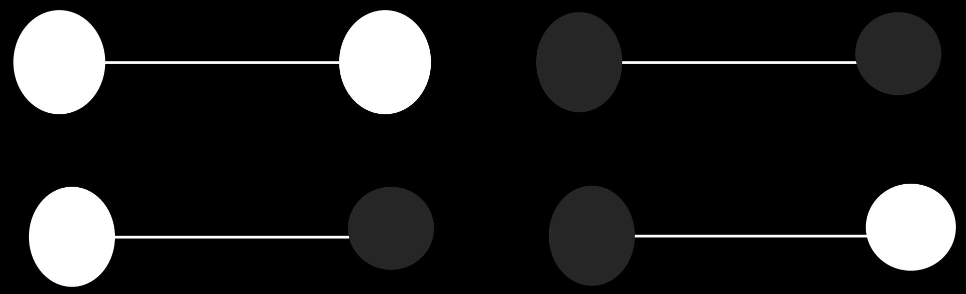
$$f(x) = \frac{x_i + x_i - 2x_i x_j}{2} \approx \frac{1 - Z_i Z_j}{2}$$

$$U_B(\beta) \approx e^{-i\beta H_B} \approx \cos(\beta)\, \mathrm{I} - \mathrm{i}\sin(\beta)\, \mathrm{X}$$

$$U_c(\gamma) \approx e^{-i\gamma H_c} \approx \cos(\tfrac{\gamma}{2})\, \mathrm{I} + \mathrm{i}\sin(\tfrac{\gamma}{2})\, Z_i Z_j$$

For p=1

$$|\Psi> = \boldsymbol{U_B(\beta)\, U_c(\gamma)}\, |\Psi_{\mathbf{0}}>$$



Corresponding Gates

$$\boldsymbol{U_B(\beta)} \approx \mathrm{R_x}(\boldsymbol{\beta})$$

$$\boldsymbol{U_c(\gamma)} \approx \mathrm{CNOT}\; R_z(-\frac{\gamma}{\mathbf{2}})\; CNOT$$

$$|\Psi_{\mathbf{0}}> = \frac{1}{2}(|00> +|01> +|11> +|10>)$$

$$U_B(\beta) = \begin{bmatrix} \cos^2\beta & -i\cos\beta\sin\beta & -i\cos\beta\sin\beta & -\sin^2\beta \\ -i\cos\beta\sin\beta & \cos^2\beta & -\sin^2\beta & -i\cos\beta\sin\beta \\ -i\cos\beta\sin\beta & -\sin^2\beta & \cos^2\beta & -i\cos\beta\sin\beta \\ -\sin^2\beta & -i\cos\beta\sin\beta & -i\cos\beta\sin\beta & \cos^2\beta \end{bmatrix}$$

$$\mathrm{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$R_z^{(2)}(-\gamma/2) = \begin{pmatrix} e^{i\gamma/2} & 0 & 0 & 0 \\ 0 & e^{-i\gamma/2} & 0 & 0 \\ 0 & 0 & e^{i\gamma/2} & 0 \\ 0 & 0 & 0 & e^{-i\gamma/2} \end{pmatrix}$$

$$U_C(\gamma) = \begin{pmatrix} e^{i\gamma/2} & 0 & 0 & 0 \\ 0 & e^{-i\gamma/2} & 0 & 0 \\ 0 & 0 & e^{-i\gamma/2} & 0 \\ 0 & 0 & 0 & e^{i\gamma/2} \end{pmatrix}$$

$$|\psi_0\rangle = \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

Analytical Approach

$$U_c(\gamma) \approx e^{(i\frac{\gamma}{2})Z_iZ_j}$$

Using cost Hamiltonian

$$Z\,|0> = +1\,|0>$$
$$Z\,|1> = -1|1>$$

INPUT		OUTPUT		Z_iZ_j	$U_c(\gamma)$
i	j	Z_i	Z_j	-	-
$\|0>$	$\|0>$	$+\|0>$	$+\|0>$	$+\|00>$	$e^{i\frac{\gamma}{2}}\,\|00>$
$\|0>$	$\|1>$	$+\|0>$	$-\|1>$	$-\|01>$	$e^{-i\frac{\gamma}{2}}\,\|01>$
$\|1>$	$\|0>$	$-\|1>$	$+\|0>$	$-\|10>$	$e^{-i\frac{\gamma}{2}}\,\|10>$
$\|1>$	$\|1>$	$-\|1>$	$-\|1>$	$\|11>$	$e^{i\frac{\gamma}{2}}\,\|11>$

$$U_c(\gamma)\,|\Psi_0> = \frac{1}{2}(e^{i\frac{\gamma}{2}}|00> + e^{-i\frac{\gamma}{2}}|01> + e^{-i\frac{\gamma}{2}}|10> + e^{i\frac{\gamma}{2}}|11>)$$

Using Matrix form of Gate

$$U_C(\gamma)\,|\psi_0\rangle \;=\; \frac{1}{2}\begin{pmatrix} e^{i\gamma/2} & 0 & 0 & 0 \\ 0 & e^{-i\gamma/2} & 0 & 0 \\ 0 & 0 & e^{-i\gamma/2} & 0 \\ 0 & 0 & 0 & e^{i\gamma/2} \end{pmatrix}\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$U_c(\gamma)\,|\Psi_0> = \frac{1}{2}(e^{i\frac{\gamma}{2}}|00> \; + e^{-i\frac{\gamma}{2}}|01> + e^{-i\frac{\gamma}{2}}|10> \; + e^{i\frac{\gamma}{2}}|11>)$$

$$|\Psi> = \boldsymbol{U_B(\beta)\,U_c(\gamma)\,|\Psi_0>}$$

$$U_M(\beta)\,U_C(\gamma)\,|\psi_0\rangle = \frac{1}{2}\Bigg[\begin{pmatrix} \cos^2\beta & -i\cos\beta\sin\beta & -i\cos\beta\sin\beta & -\sin^2\beta \\ -i\cos\beta\sin\beta & \cos^2\beta & -\sin^2\beta & -i\cos\beta\sin\beta \\ -i\cos\beta\sin\beta & -\sin^2\beta & \cos^2\beta & -i\cos\beta\sin\beta \\ -\sin^2\beta & -i\cos\beta\sin\beta & -i\cos\beta\sin\beta & \cos^2\beta \end{pmatrix}$$

$$\times \begin{pmatrix} e^{i\gamma/2} & 0 & 0 & 0 \\ 0 & e^{-i\gamma/2} & 0 & 0 \\ 0 & 0 & e^{-i\gamma/2} & 0 \\ 0 & 0 & 0 & e^{i\gamma/2} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}\Bigg]$$

$$|\psi\rangle = \frac{1}{2}\begin{pmatrix} e^{i\gamma/2}\cos 2\beta - i\,e^{-i\gamma/2}\sin 2\beta \\ -i\,e^{i\gamma/2}\sin 2\beta + e^{-i\gamma/2}\cos 2\beta \\ -i\,e^{i\gamma/2}\sin 2\beta + e^{-i\gamma/2}\cos 2\beta \\ e^{i\gamma/2}\cos 2\beta - i\,e^{-i\gamma/2}\sin 2\beta \end{pmatrix}$$

$$|\psi\rangle = \frac{1}{2}\begin{bmatrix} \cos\left(2\beta + \frac{\gamma}{2}\right) + i\sin\left(2\beta - \frac{\gamma}{2}\right) \\ \cos\left(2\beta - \frac{\gamma}{2}\right) - i\sin\left(2\beta + \frac{\gamma}{2}\right) \\ \cos\left(2\beta - \frac{\gamma}{2}\right) - i\sin\left(2\beta + \frac{\gamma}{2}\right) \\ \cos\left(2\beta + \frac{\gamma}{2}\right) + i\sin\left(2\beta - \frac{\gamma}{2}\right) \end{bmatrix}$$

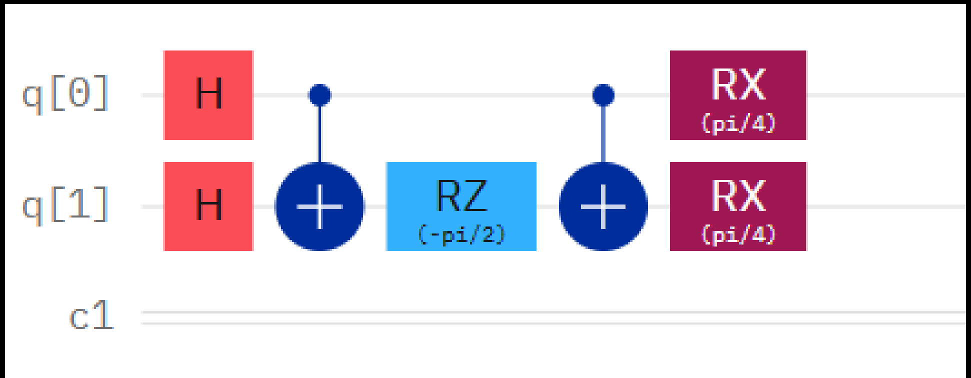
$$C(\beta,\gamma) = <\psi_Q(\beta,\gamma)|H_C|\psi_Q(\beta,\gamma)>$$

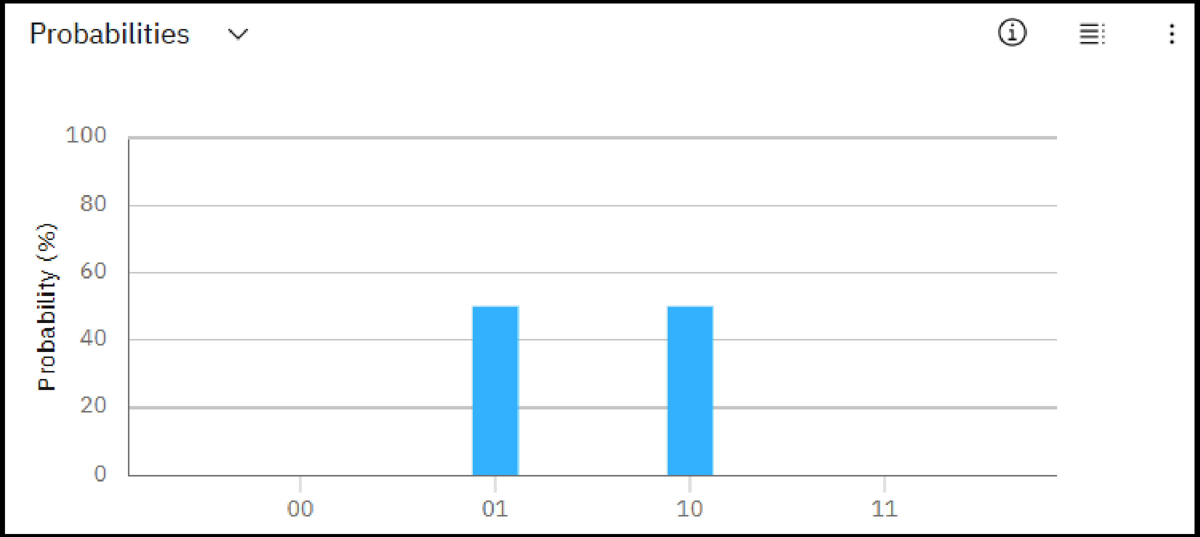
$$H_C = \frac{1}{2} - \frac{Z_i Z_j}{2}$$

$$C(\beta,\gamma) = \frac{1}{2} + \frac{1}{2}(\sin 4\beta \sin\gamma)$$

$$(\beta,\gamma) = ((2n+1)\frac{\pi}{8}, (2n+1)\frac{\pi}{2})$$

Gate Level implementation





QAOA for General Graph

Then for each edge (u ,v) of the graph,

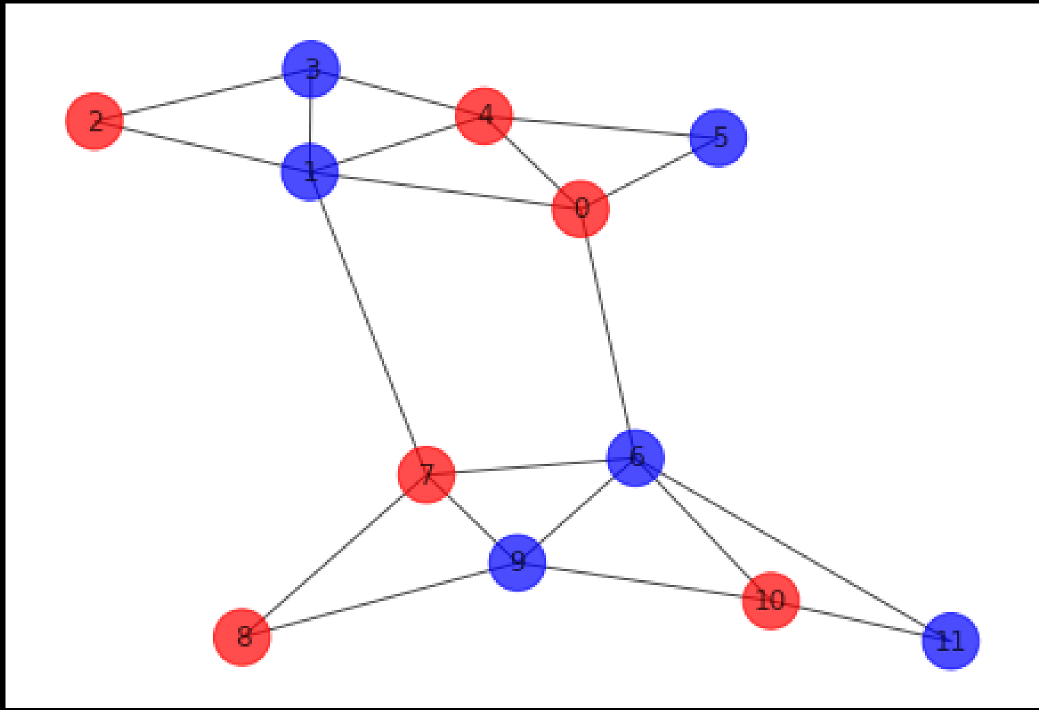
$$< \psi(\beta,\gamma)|C_{uv}|\psi(\beta,\gamma) > = \frac{1}{2} + \frac{1}{4}\sin(4\beta)\, cos\gamma\left(cos^{d}\gamma + cos^{e}\gamma\right) - \frac{1}{4}sin^{2}(2\beta)cos^{d+e-2f}\,\gamma(1 - cos^{f}(2\gamma))$$

d = deg(u) − 1, e = deg(v) − 1,

f is the number of triangles in the graph containing (u ,v).

The overall expectation value of graph is

$$< C > = < C_{uv} > \times No\ of\ edges(d, e, f)$$



Triangle-free Graphs

D-regular triangle-free graph,

$$\langle C \rangle = \frac{m}{2} + \frac{m}{2}\sin(4\beta)\sin(\gamma)\cos^{D-1}(\gamma)$$

$$\langle \boldsymbol{R} \rangle = \frac{\langle \boldsymbol{C} \rangle}{\boldsymbol{m}}\,, \qquad m\ is\ total\ no.of\ edges$$

$$\max_{\gamma,\beta}\langle R \rangle \geq \frac{1}{2} + \frac{1}{2\sqrt{e}}\frac{1}{\sqrt{D}}$$

max<R> $\geq$ 0.75, 0.69245, 0.66238, 0.64310

for D = 2, 3, 4, 5

Any arbitrary triangle-free graph,

$$\langle C \rangle = \frac{m}{2} + \frac{1}{4}\sin(4\beta)\sin(\gamma)\sum_{D} D\, n_D \cos^{D-1}(\gamma)$$

$$\max_{\gamma,\beta}\langle R \rangle \geq \frac{1}{2} + \frac{1}{2\sqrt{e}}\frac{1}{\sqrt{D_G}}$$

Here D_G is maximum degree

Constraint Optimization

Soft Constraints

- **These constraints should be respected.**
- **Enforced by penalties in the cost Hamiltonian.**

Hard Constraints

- **These constraints must be respected.**
- **Infeasible states are never explored by the mixer Hamiltonian.**

Variants of QAOA (for constraint optimization)

❑ *QAOA+*

"QAOA+ is standard QAOA with enhanced mixers"

$$U_{\text{mixer}}^{\text{QAOA+}} = e^{-i\beta H_M^{\text{enhanced}}}$$

❑ *RQAOA*

"RQAOA solves large problems by recursively fixing highly biased qubits after each QAOA run, reducing problem size."

Future Goals

"Develop quantum algorithms using enhanced and recursive QAOA for real-life constrained optimization."

Summary

- Explain what combinatorial optimization problems are and why they are computationally hard.
- Study quantum algorithms designed to solve these problems, and identify QAOA as a strong candidate .
- Analyze the Max-Cut problem for both simple and general graphs, and derive a general formula for the expected cost .
- Compute the approximation ratio and study how it changes with graph degree.
- Explain what constraints are in optimization problems and how quantum algorithms can handle them.
- Future direction: Extend this work to constrained optimization problems and develop quantum algorithms for real-world applications.

References

- *https://www.itp.unihannover.de/fileadmin/itp/qinfo/Team_Tobias_Osborne/Bachelors_Theses/Jan_Rasmus_Holst_Bachelors_Thesis.pdf*
- *https://arxiv.org/pdf/2004.09002*
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THANK YOU